✅ Loan Default Risk Analysis and Prediction

Capstone Project

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1. Project Title

Loan Default Risk Analysis and Prediction

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2. Business Problem / Background

Financial institutions such as banks and NBFCs provide loans to customers based on income, credit history, and personal details. One of the biggest challenges faced by these institutions is **loan default**, where customers fail to repay loans on time.

Loan defaults result in:

- Financial losses
- Increased Non-Performing Assets (NPAs)
- Higher operational and recovery costs

To minimize risk, banks aim to **analyze historical loan data** and **predict default probability** before approving loans.

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3. Problem Statement

Analyze historical loan applicant data to identify key factors influencing loan default and build a machine learning model to predict whether a loan applicant is likely to default.

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4. Project Objectives

1. Perform **Exploratory Data Analysis (EDA)** using univariate and bivariate techniques
2. Understand relationships between customer attributes and loan default
3. Properly handle numeric, categorical, and binary variables
4. Build a predictive classification model for loan default risk
5. Provide **data-driven insights** to reduce future defaults

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5. Dataset Description

✅ Dataset Source

The **Loan Prediction Dataset** (Kaggle / Analytics Vidhya), widely used for academic projects and interviews.

✅ Important Features

Column Name	Description	Data Type
Loan_ID	Unique loan identifier	Categorical (ID)
Gender	Applicant gender	Categorical
Married	Marital status	Categorical
Dependents	Number of dependents	Categorical
Education	Graduate / Not Graduate	Categorical
Self_Employed	Employment type	Categorical
ApplicantIncome	Applicant income	Numeric
CoapplicantIncome	Co-applicant income	Numeric
LoanAmount	Loan amount requested	Numeric
Loan_Amount_Term	Loan repayment term	Numeric
Credit_History	Credit history availability	Binary (0/1)
Property_Area	Urban / Semiurban / Rural	Categorical
Loan_Status	Loan approved or defaulted	Target

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6. Folder-Wise Project Execution

📁 1. Data Collection

- Load raw loan dataset (CSV)
- Initial understanding of data
  ✅ Output: Raw dataset

📁 **2. Data Pre-Processing**

- Handle missing values
- Convert Loan_ID to string
- Encode categorical variables
- Convert Loan_Status to binary (0/1)
- Separate numeric and categorical variables

✅ Output: Cleaned, model-ready dataset

📁 **3. Univariate & Bivariate Analysis**

✅ **Univariate Analysis**

- Distribution of LoanAmount
- Mean & median of ApplicantIncome
- Frequency of Loan_Status
- Applicant count by education & property area

✅ **Bivariate Analysis**

- Income vs Loan Status
- Credit History vs Loan Status
- Property Area vs Default
- Loan Amount vs Default Probability

⚠️ Only numeric columns used for:

- Mean
- Median
- Quartiles

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7. Feature Selection & Model Creation

- Train-test split
- Feature importance analysis

- Algorithms used:
    - Logistic Regression
    - Decision Tree
    - Random Forest

✅ **Evaluation Metrics**

- Accuracy
- Precision
- Recall
- Confusion Matrix

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8. Final Model

- Best-performing model selected
- Final evaluation metrics
- Business-oriented conclusions
- Recommendations to the bank

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9. Key Business Questions Answered

- Does credit history strongly influence loan default?
- Do higher loan amounts increase default risk?
- Is income alone sufficient for approval decisions?
- Which features are most important in predicting default?

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10. Expected Outcomes

- Identify high-risk loan applicants
- Reduce NPAs using data-driven decisions
- Improve loan approval efficiency
- Support banks in risk assessment

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11. Interview Explanation (Short & Effective)

"I worked on a Loan Default Risk Analysis project where I performed EDA to understand customer behavior and built a classification model to predict loan default using historical banking data."

✅ Clear
✅ Professional
✅ Confident

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12. Variable Classification

Type	Variable Name	Role
Dependent	Loan_Status	Outcome
Independent	Gender	Input
	Married	Input
	Dependents	Input
	Education	Input
	Self_Employed	Input
	ApplicantIncome	Input
	CoapplicantIncome	Input
	LoanAmount	Input
	Loan_Amount_Term	Input
	Property_Area	Input
	Credit_History	Input

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13. Key Insights from Analysis

✅ **Strongest Predictors**

- **Credit History**
  Applicants with credit history are significantly less likely to default.

- **Income (Applicant + Co-Applicant)**
  Higher combined income reduces default risk.
- **Loan Amount**
  Larger loan amounts relative to income increase default probability.

⚖️ **Moderate Predictors**

- Loan Amount Term
- Education Level
- Property Area

🟡 **Weaker / Contextual Factors**

- Dependents
- Marital Status
- Gender
- Self-Employed status (context dependent)

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14. Machine Learning Approach

a) Pre-Processing

- One-hot encoding for categorical features
- Scaling numeric features
- Handling missing values

b) Feature Engineering

- Total household income
- Loan-to-income ratio
- Loan term buckets

c) Model Selection

- Logistic Regression
- Random Forest

- Gradient Boosting / XGBoost
- SVM (optional)

d) Evaluation

- Accuracy, Precision, Recall
- F1-Score
- ROC-AUC
- K-Fold Cross Validation

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15. Final Outcome

The final model provides:

- **Probability of default**
- **Risk classification (Default / Non-Default)**
- **Feature importance for business transparency**

✅ **Final Recommendation**

This project is strong because it:

✓ Solves a real business problem
✓ Uses real-world data
✓ Covers the full Data Science lifecycle
✓ Demonstrates strong analytical & ML skills